

Airbnb Price Prediction Report - New York City

Generated on 2026-08-25 by the automated pipeline (*make all*).

Executive summary

The pipeline trained and compared Random Forest, XGBoost and TabPFN on Airbnb listing data for **New York City** (14929 listings). The **xgboost** model was selected for final evaluation on a held-out test set and achieved: - **RMSE: \$572** - **MAE: \$116** - **R²: 0.92**

Data

- Source: InsideAirbnb (latest available snapshot). See: <https://insideairbnb.com/get-the-data/>
- City: New York City · 14929 listings after cleaning

EDA

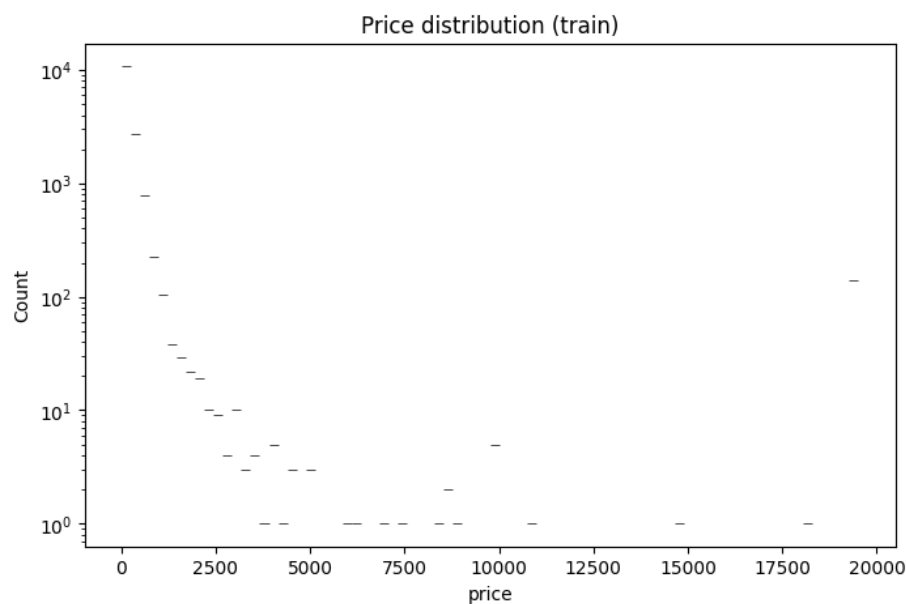
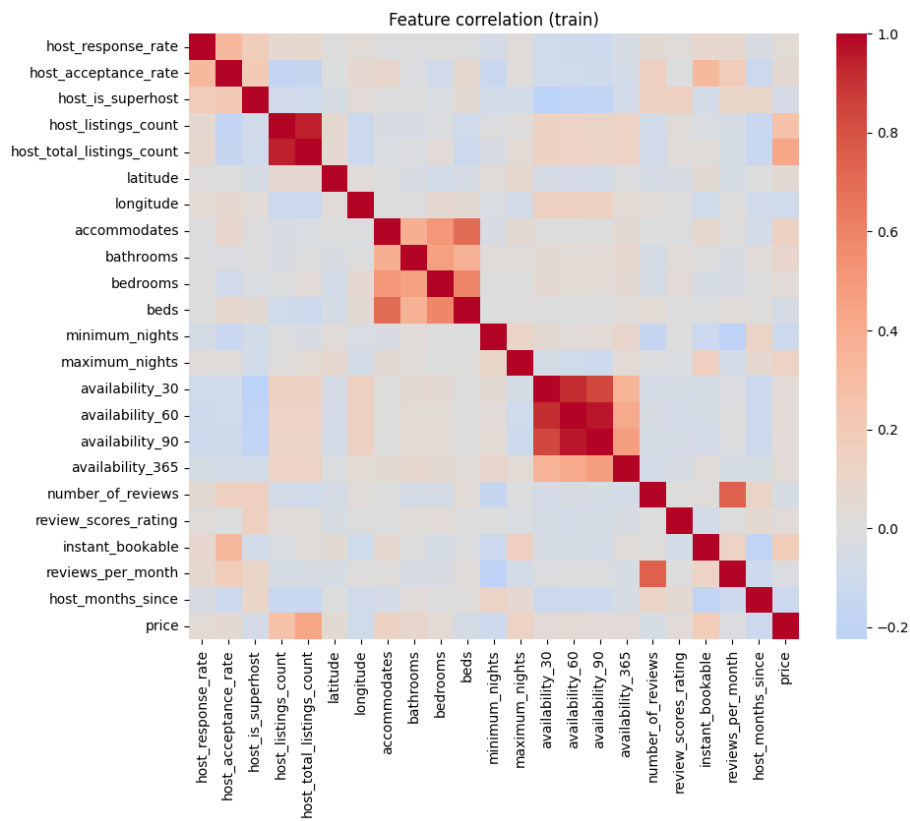
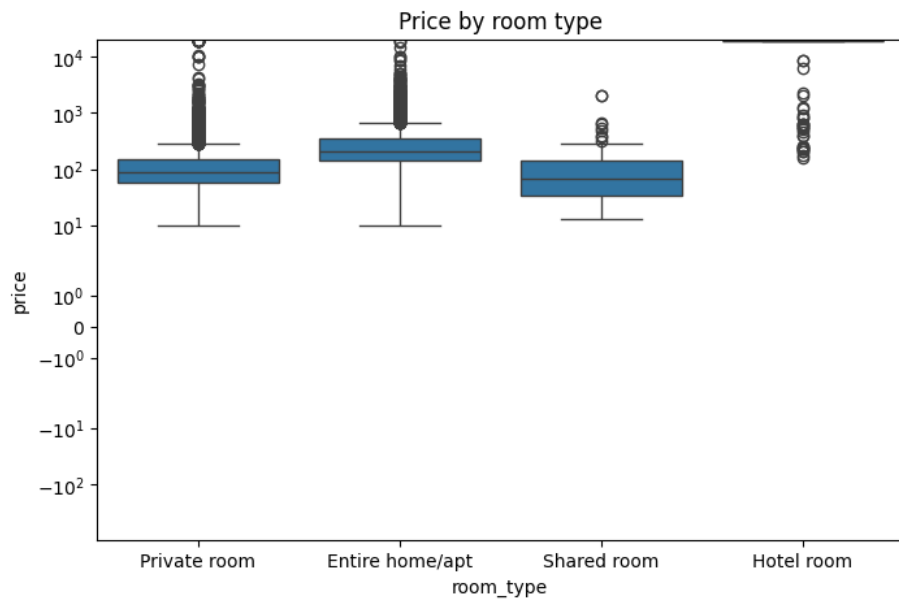


Figure 1: price_dist

- Median price: **\$154** (mean \$420)



Models & Results

Model	Validation RMSE
Random Forest	\$713
XGBoost	\$667
TabPFN (capped sample, not directly comparable)	\$956

Holdout results (best model): see `artifacts/new-york-city/holdout_results.csv`.

Top features

Feature	Importance
room_type_Hotel room	0.400
host_response_time_within an hour	0.134
host_neighbourhood_Gateway District	0.084
host_neighbourhood_Central Business District	0.049
host_months_since	0.027
host_neighbourhood_Airport North	0.025
property_type_Room in hotel	0.022
beds	0.021
host_neighbourhood_Clearwater Beach	0.019
minimum_nights	0.018

Recommendations

- The best-performing model was **xgboost** (holdout RMSE \$572, MAE \$116, R^2 0.92).
- On validation, **XGBoost** beat Random Forest by 6% on RMSE (\$667 vs \$713). A simple ensemble or blending the two may squeeze out further gains.
- Pricing is most strongly driven by **room_type_Hotel room**, **host_response_time_within an hour**, **host_neighbourhood_Gateway District** — i.e. room_type_Hotel room (the room type (entire home vs private/shared room) (specifically Hotel room)); host_response_time_within an hour (how quickly the host responds (specifically within an hour)); host_neighbourhood_Gateway District (the host's neighbourhood (specifically Gateway District)). Hosts can use these levers when setting nightly rates.
- Accuracy caveat: the holdout RMSE of \$572 is **371% of the median listing price (\$154)**, so predictions for typical listings are expected to be off by roughly $\pm \$572$.

- Next steps: add richer features (neighbourhood_cleansed, amenities, review scores breakdowns), train on a log-transformed target to tame high-end outliers, run a deeper hyperparameter search, and extend the pipeline to more cities.

*Automated pipeline: fetch → preprocess → EDA → train → evaluate → report.
Code in **src/**, config in **configs/base.yaml**.*